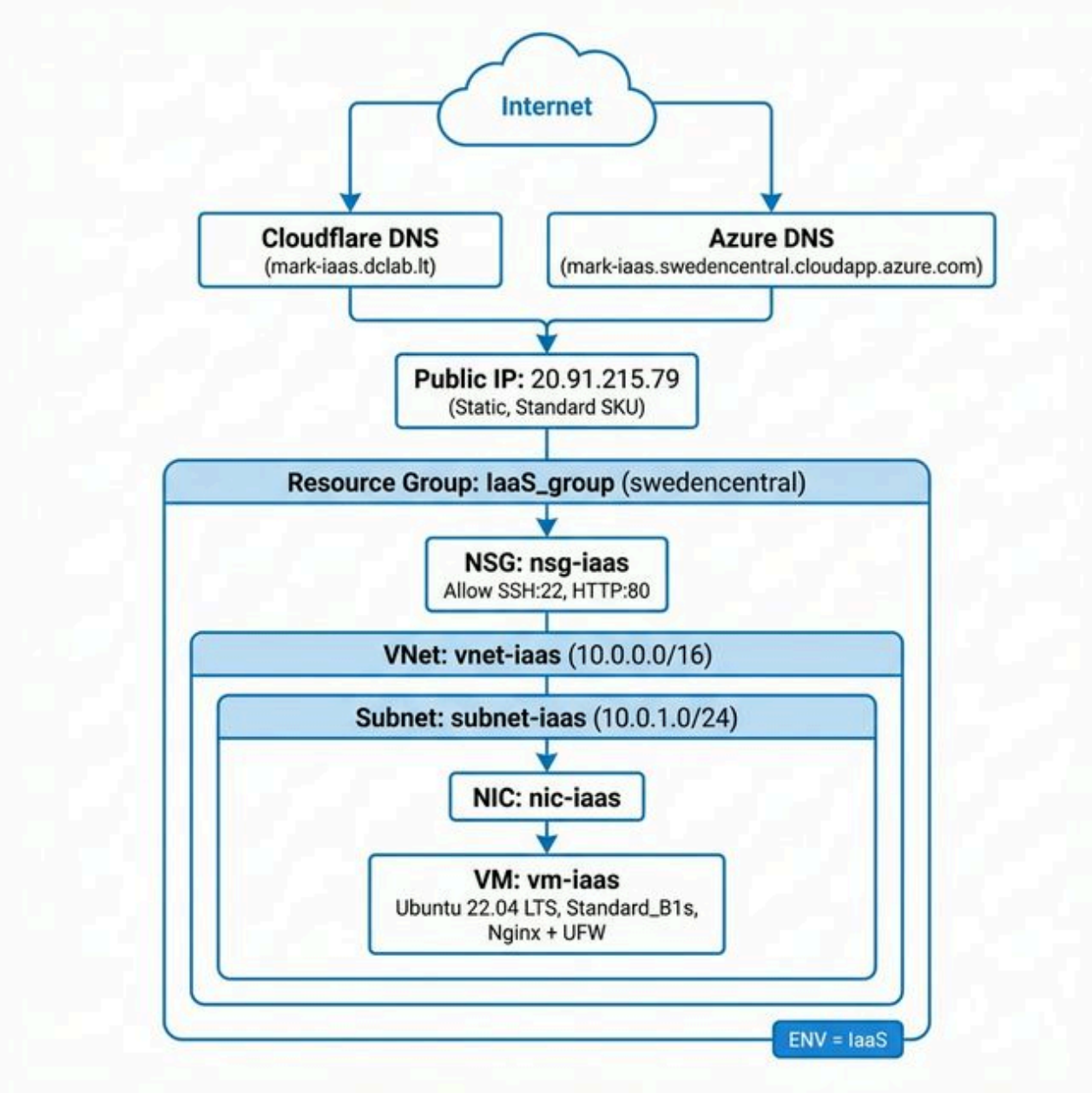
1. Objective

Create IaaS resources on the Azure cloud computing platform using Infrastructure as Code (OpenTofu) to provision a virtual machine running Ubuntu 22.04 with Nginx web server.

2. Architecture



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3. Work Progress

3.1 Environment Setup

- Authenticated with Azure CLI (`az login`)
- Initialised an OpenTofu project with the `azurerm` provider

3.2 Resource Group

- Created resource group **IaaS_group** in the **swedencentral** region
- Initially attempted `eastus` , but the subscription policy only allows: `polandcentral` , `uksouth` , `italynorth` , `swedencentral` , `germanywestcentral`
- Applied the tag `ENV = IaaS` to all resources

3.3 Networking

- **Virtual Network** `vnet-iaas` with address space `10.0.0.0/16`
- **Subnet** `subnet-iaas` with prefix `10.0.1.0/24`
- **Public IP** `pip-iaas` — Static allocation, Standard SKU
  - DNS label: `mark-iaas.swedencentral.cloudapp.azure.com`
  - IP address: `20.91.215.79`
- **Network Security Group** `nsg-iaas` — allows inbound TCP on ports **22** (SSH) and **80** (HTTP)
- NSG associated to both the subnet and the NIC

3.4 Virtual Machine

Property	Value
Name	`vm-iaas`
OS	Ubuntu 22.04 LTS (Canonical)
Size	`Standard_B1s` (1 vCPU, 1 GB RAM)
Admin user	`azureuser`
Authentication	SSH key + password
OS disk	Standard_LRS

3.5 Cloud-Init Provisioning

On first boot, **cloud-init** automatically:

1. Updated all packages
2. Installed **Nginx** web server
3. Installed and configured **UFW** firewall (allow ports 22, 80)
4. Deployed a custom VilniusTech-styled `index.html` to `/var/www/html/`

3.6 Connectivity Testing

- Accessed the web page via Public IP: `http://20.91.215.79` ✅
- Accessed the web page via Azure DNS: `http://mark-iaas.swedencentral.cloudapp.azure.com` ✅
- Connected via SSH: `ssh -i id_rsa azureuser@mark-iaas.swedencentral.cloudapp.azure.com` ✅

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3.7 Cloudflare DNS

- Created an **A record** in Cloudflare for `mark-iaas.dclab.lt` → `20.91.215.79`
- Verified connectivity via the custom domain ✅

3.8 Cleanup

- All resources destroyed with `tofu destroy`

4. Resource Group Overview

All resources under **IaaS_group** tagged with `ENV = IaaS`:

#	Resource	Type	Region
1	`vnet-iaas`	Virtual Network	swedencentral
2	`subnet-iaas`	Subnet	swedencentral
3	`pip-iaas`	Public IP Address	swedencentral
4	`nsg-iaas`	Network Security Group	swedencentral
5	`nic-iaas`	Network Interface	swedencentral
6	`vm-iaas`	Linux Virtual Machine	swedencentral

5. Questions

What were the main steps involved in creating and configuring virtual machines in Azure?

1. **Resource Group creation** — a logical container for all related resources
2. **Networking setup** — Virtual Network, Subnet, Public IP, and Network Security Group define how the VM connects to the internet and what traffic is permitted
3. **Network Interface** — bridges the VM to the subnet and public IP
4. **VM provisioning** — selecting the OS image (Ubuntu 22.04), size (Standard_B1s), and authentication method (SSH key + password)
5. **Post-boot configuration** — cloud-init installed Nginx, configured UFW firewall, and deployed the web page automatically on first boot

Did you encounter any challenges during the lab? How did you overcome them?

- **Azure region policy (403 Forbidden):** The subscription restricted resource deployment to specific regions. Initially used `eastus`, which was blocked. Queried the Azure policy with `az policy assignment list` and discovered only 5 regions were allowed. Switched to `swedencentral`.
- **Missing provider declarations:** The `tls` and `local` providers for SSH key generation were used but not declared, causing init warnings. Added explicit declarations in `versions.tf`.

How did you connect to your virtual machine? What protocols did you use, and why?

SSH (port 22) was used for remote terminal access to administer the VM. SSH provides encrypted communication and supports both key-based and password-based authentication. Two methods were available:

- **SSH key:** `ssh -i id_rsa azureuser@<DNS>` — more secure, uses a 4096-bit RSA key pair generated by OpenTofu
- **Password:** `ssh azureuser@<DNS>` — simpler but less secure; enabled as a lab requirement

How did you ensure the security of your virtual machines?

1. **Network Security Group** — acts as a firewall, allowing only SSH (22) and HTTP (80) traffic; all other ports are denied by default
2. **UFW firewall** — configured on the VM itself as a second layer of defense
3. **SSH key authentication** — 4096-bit RSA key pair generated automatically, private key stored locally with `0600` permissions
4. **No public inbound access** except the two required ports
5. **NSG associated to both NIC and subnet** — defense in depth

Explain different ways to connect to your hosted site

1. **Public IP address:** `http://20.91.215.79` — direct access using the Static IP assigned to the VM
2. **Azure DNS label:** `http://mark-iaas.swedencentral.cloudapp.azure.com` — Azure-managed FQDN configured on the Public IP resource
3. **Custom domain (Cloudflare):** `http://mark-iaas.dclab.lt` — a DNS A record in Cloudflare pointing the custom domain to the VM's public IP